

Lecture 01 : HTML Basics

Hyper Text Markup Language | First Web Page, Essential Tags & Mini Project

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1.0 Overview

HTML (Hyper Text Markup Language) web development ka sabse pehla aur sabse important step hai. Is lecture mein hum duniya ki pehli website se lekar apna khud ka first web page aur ek complete mini project tak ka safar tay karenge. Bina kisi prior knowledge ke, step-by-step.

2.0 HTML Kya Hai?

PATTERN

HTML = **Hyper Text** + **Markup Language**

Hyper Text matlab aisa text jo ek page se doosre page tak navigate kar sake (hyperlinks ke through). Markup Language matlab aisi language jo content ko **semantic meaning** aur **structure** deti hai.

2.1 Hyper Text ka Matlab

Duniya ki pehli website **World Wide Web** thi. Usme koi design nahi thi — sirf plain text aur links the. Agar aap kisi link pe click karte the, toh aap doosre page pe redirect ho jaate the. Inhi links ko hum **Hyperlinks** kehte hain.

INTUITION

Socho ek aisi library jisme koi librarian nahi hai. Har book ke andar chitthi hai jo batati hai ki agli book kahan hai. Aap ek se doosri, doosri se teesri book tak manually navigate karte ho. Yehi concept hyperlinks ka hai — ek page se doosre page tak manual navigation.

Us time pe browsers mein search functionality nahi hoti thi. Toh agar aapko kuch dhoondhna tha, toh aapko manually ek-ek karke links follow karne padte the. Aaj hum Google pe search karke direct result pa lete hain, par pehle aisa nahi tha.

2.2 Markup Language ka Matlab

Ab yeh samajhna important hai ki **Markup** ka matlab kya hai. Jab aap bina tags ke plain text likhte ho, toh browser sirf ek hi line mein sab kuch dikhata hai. Lekin jab aap `tags` ka use karte ho, toh aap browser ko batate ho ki "yeh heading hai", "yeh paragraph hai", "yeh list hai". Isi ko Markup Language kehte hain.

3.0 VS Code Setup

Code likhne ke liye humein ek **Code Editor** chahiye. Hum **VS Code** use karenge kyunki ye free hai, powerful hai, aur extensions ki madad se aur bhi zyada useful ban jaata hai.

WORKFLOW

Step-by-Step VS Code Setup:

1. VS Code official website se download karo (Windows/Mac/Linux ke liye automatic detect ho jaayega)
2. Install karo system mein
3. Extensions tab mein jaake teen extensions install karo
4. Ek naya folder create karo (jaise "Defense")
5. VS Code mein **Open Folder** se us folder ko khol lo
6. Nayi file banao: `index.html`

3.1 Essential Extensions

Ye teen extensions aapko har baar install karne hain:

Extension Name	Kaam Kya Hai
Live Server	Browser mein automatically page refresh karta hai jab bhi code save karte ho
Live Preview	VS Code ke andar hi side panel mein website ka preview dikhata hai
Prettier	Code ko automatically format aur beautify karta hai

CONFIG

Auto Save Enable Karo: VS Code mein *File* → *Auto Save* ko on kar lo. Isse baar-baar Ctrl+S dabane ki zaroorat nahi padegi. Code likhte hi automatic save ho jaayega.

4.0 Pehla Web Page

4.1 index.html Kyon?

Jab aap koi website khulte ho, toh server sabse pehle `index.html` file ko dhoondhta hai. Ye default file hoti hai. Isliye hamesha pehli file ka naam **index.html** rakho.

4.2 File Extension ka Concept

Har file ka ek **extension** hota hai jo batata hai ki system ko us file ko kaise read karna hai:

File Type	Extension	System Kya Samajhta Hai
Video	.mp4	Ye video file hai
Image	.jpg / .png	Ye image file hai
Web Page	.html	Ye HTML document hai — browser isko render karega

4.3 First Code

Ab apna pehla web page banate hain. Bas itna likho:

```
Hello Coder Army! Kaise hain aap sab log?
```

Is file ko save karo, phir **Live Server** se open karo. Aapka pehla web page ban chuka hai! Itna aasan tha.

DEPLOYMENT

Live Server ka use kaise karein: File pe right-click karo → *Open with Live Server* select karo. Aapka default browser mein ek naya tab open hoga aur aapka page dikhega. Jab bhi code change karoge, page automatically refresh hoga.

5.0 Tags ka Concept

Ab ek important observation notice karo. Agar aap plain text likhte ho:

```
Web Development Course
In this course I will teach students how to become full stack developer from zero to advance.
```

Toh browser mein ye ek hi line mein dikhega, chahe aapne kitni bhi spaces diye ho. Browser extra spaces ko ignore kar deta hai. Isliye humein **tags** ki zaroorat padti hai.

5.1 Tags Kya Hote Hain?

Tags **semantic meaning** dete hain content ko. Matlab browser ko pata chal jaata hai ki "yeh kya cheez hai". Jaise:

```
<h1>Web Development Course</h1>
<p>In this course I will teach students how to become full stack developer from zero to
advance.</p>
```

Ab browser samajh gaya ki **"Web Development Course"** ek heading hai aur neeche wala content ek paragraph hai. Ye structure create karna hi Markup Language ka kaam hai.

6.0 Heading Tags (H1 – H6)

IMPORTANT

HTML mein total **6 types** ki headings hoti hain: H1, H2, H3, H4, H5, H6

Inka kaam sirf text size badana **NAHI** hai. Ye **semantic priority** batati hain ki kaunsa content kitna important hai.

6.1 Semantic Priority Samajhna

Dhyan se dekho yeh line — **H1** ke andar jo content hai, uski priority sabse high hoti hai. Ye poore web page ka sabse important heading hai. **H2** uski sub-heading hai, **H3** uski bhi sub-heading, aur aise hi **H6** tak jaata hai.

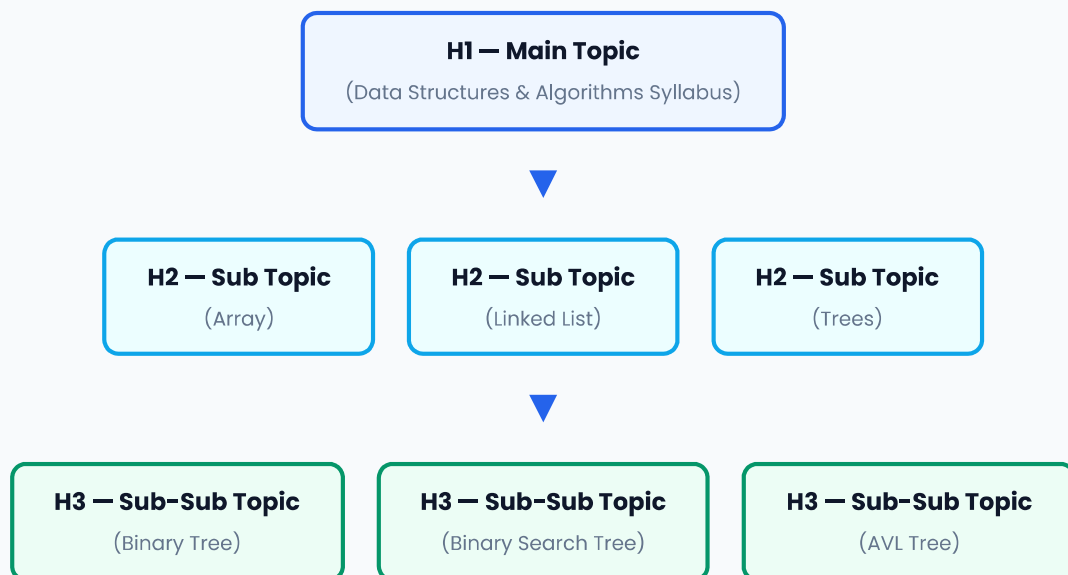
WARNING

Common Mistake: Students sochte hain ki H1 sabse bada dikhta hai isliye H1 important hai. **Galat!** CSS se aap kisi bhi heading ka size change kar sakte ho. H1 important hai kyunki **search engines** isko sabse pehle padhte hain. Font size se koi matlab nahi.

6.2 Heading Hierarchy

Ek page mein heading hierarchy hamesha maintain karni chahiye. Kabhi bhi jump mat karo:

Heading Hierarchy Tree



H1 → H2 → H3 → H4 is the correct flow. H1 → H3 jump mat karo.

6.3 SEO mein Kaise Help Karti Hain?

Search engines (jaise Google) aapki poori website read nahi karte. Wo sabse pehle **headings** ko read karte hain taaki samajh sake ki is page mein kya content hai.

INTERVIEW TIP

SEO (Search Engine Optimization) mein headings ka bahut bada role hai. Google bot sabse pehle H1 heading padhta hai — agar usme "Data Structures & Algorithms Syllabus" likha hai, toh Google samajh jaata hai ki ye page DSA ke baare mein hai. Phir wo H2 headings ko check karta hai ki sub-topics kya hain. Isi liye **semantic heading structure** follow karna critical hai.

IMPORTANT

Golden Rule: Ek page mein sirf **ONE H1** heading honi chahiye.

Multiple H1 daalne se error nahi aayega — HTML ki khasiyat hai ki galat code bhi chal jaata hai. Lekin ye **recommended nahi** hai. SEO aur accessibility ke liye hamesha ek hi H1 rakho.

6.4 Code Example

```
⚡— Correct Heading Structure —>
<h1>Data Structures & Algorithms Syllabus</h1>

<h2>Array</h2>
<p>Array is a collection of elements ... </p>

<h2>Linked List</h2>
<p>Linked List is a linear data structure ... </p>

<h2>Trees</h2>
<h3>Binary Tree</h3>
<h3>Binary Search Tree</h3>
```

7.0 Paragraph & Formatting Tags

7.1 Paragraph Tag (P)

Normal content ke liye hum `<p>` tag ka use karte hain. Jo bhi text paragraphs mein likhna hai, usko `<p>` ke andar rakho.

```
<p>This course is for Web Development.</p>
<p>Here we will discuss a lot of things.</p>
```

7.2 Lorem Ipsum (Dummy Text)

Kai baar humein dummy text chahiye hota hai design test karne ke liye. VS Code mein bas `lorem50` likho aur Enter dabao — 50 words ka random text aa jaayega. `lorem10` se 10 words, `lorem100` se 100 words.

CREATIVE TIP

Lorem Ipsum ka use tab karo jab aapko quickly layout test karna ho aur actual content abhi tak ready nahi hai. Ye Latin text hota hai jo readable nahi hota, isliye user ka dhyan sirf design pe rehta hai.

7.3 Horizontal Rule (HR)

Agar aapko ek horizontal line draw karni hai content ke beech mein, toh `<hr>` tag ka use karo. Ye ek **self-closing tag** hai — iska closing tag nahi hota.

```
<hr /> ← Recommended method with slash →  
← Ya phir aise bhi chalega: <hr> →
```

7.4 Break Tag (BR)

Agar aapko ek line ke andar hi line break chahiye (jaise poem likhni ho), toh `<br>` tag ka use karo. Ye bhi self-closing tag hai.

```
<p>Line one<br /><br />Line two</p>
```

PERFORMANCE

Self-Closing Tags ka Recommended Method: Hamesha slash ke saath likho — `<hr />` aur `<br />`.
Ye modern HTML5 mein best practice hai aur code ko clean dikhaata hai.

8.0 Tag vs Element

Students usually mistake kar dete hain — Tag aur Element ko same samajh lete hain. Lekin inme difference hai:

Tag	Element
Sirf markup symbol hai	Poora unit hai — start tag + content + end tag
Example: <code><h1></code> ya <code></h1></code>	Example: <code><h1>Hello World</h1></code>
Content ko surround nahi karta	Content ko surround karta hai

TAKEAWAY

Tag = `<h1>` | Element = `<h1>Content</h1>`

Jab bhi aap start tag, content, aur end tag ko milate ho, wo ban jaata hai ek **HTML Element**. Tag sirf angle brackets ke andar ka naam hai.

9.0 Lists (OL & UL)

Website mein list banana bahut common hai — jaise ingredients ki list, steps ki list, skills ki list. HTML mein do types ki lists hoti hain:

9.1 Ordered List (OL)

Jab aapko items ko **numbering** ke saath dikhana ho (1, 2, 3...), toh `<ol>` use karo. Har item ko `<li>` (List Item) tag ke andar rakho.

```
<ol>
  <li>Milk</li>
  <li>Sugar</li>
  <li>Tea Leaves</li>
  <li>Water</li>
</ol>
```

Output mein automatically 1, 2, 3, 4 numbering aa jaayegi.

9.2 Unordered List (UL)

Jab aapko items ko **bullets/dots** ke saath dikhana ho, toh `<ul>` use karo. Items same `<li>` tag ke andar jaate hain.

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Output mein dots (•) dikhenge instead of numbers.

ORDERED LIST

1. Milk
2. Sugar
3. Tea Leaves
4. Water

UNORDERED LIST

- HTML
- CSS
- JavaScript
- TypeScript

10.0 Anchor Tag & Attributes

10.1 Anchor Tag (A)

Hyperlinks create karne ke liye hum `<a>` tag ka use karte hain. Iska main attribute hota hai `href` (Hypertext Reference) — jisme aap URL daalte ho.

```
<a href="https://coderarmy.in">Visit Coder Army</a>
```

10.2 Target Attribute

By default, link **same tab** mein open hota hai. Agar aapko **new tab** mein open karna hai, toh `target="_blank"` attribute add karo.

```
<a href="https://coderarmy.in" target="_blank">Visit Coder Army</a>
```

10.3 Attributes Kya Hote Hain?

Attributes wo properties hain jo batati hain ki tag kaise behave karega. Ye hamesha **opening tag** ke andar likhe jaate hain `name="value"` format mein.

Common Attributes:**href** = Link ka URL**target** = Kahan open karna hai (_blank, _self)**src** = Image/source ka URL**alt** = Alternate text**style** = Inline CSS styles

11.0 Image Tag

Website mein images add karne ke liye `<img>` tag ka use karte hain. Ye bhi ek self-closing tag hai.

11.1 Basic Syntax

```

```

11.2 Attributes Explained

Attribute	Kaam Kya Hai	Required?
src	Image ka source URL batata hai	Yes
alt	Image fail hone par text dikhata hai; screen readers ke liye bhi important	Recommended
height	Image ki height pixels mein	Optional
width	Image ki width pixels mein	Optional

EDGE CASE

Image Load Nahi Hoti? Agar src URL galat hai ya server down hai, toh image nahi dikhegi. Us case mein **alt** attribute ka text dikh jaata hai. Ye sirf visual users ke liye nahi, **screen readers** (jo blind users use karte hain) ke liye bhi critical hai — wo image nahi dekh sakte, par alt text padh sakte hain.

BEST PRACTICE

Alt Text Hamesha Do: Accessibility guidelines ke hisaab se har image ka alt attribute hona chahiye. Agar image sirf decorative hai, toh blank alt="" rakh sakte ho. Lekin informative images ke liye meaningful description likho.

12.0 Comments

Comments wo lines hoti hain jo browser read karta hai lekin output mein **nahi dikhati**. Ye sirf developer ke liye hoti hain taaki code samajhne mein aasaan rahe.


```
⚡— Ye ek comment hai —>
⚡— Ye output mein kabhi nahi dikhega —>
⚡— Shortcut: Ctrl + / (Windows) ya Command + / (Mac) —>
```

BEST PRACTICE

Complex code sections ke upar comments likhne ki aadat daalo. 2 hafte baad jab wapas code padhoge, toh khud hi thank you bologe.

13.0 Mini Project: Front End Developer Page

Ab humne jo kuch bhi seekha, usse ek complete page banate hain. Is page mein hoga:

- H1 Heading
- Image
- Horizontal Line
- H2 Heading + Paragraphs
- H2 + Unordered List (Skills)
- H2 + Ordered List (Steps)
- Anchor Tag with New Tab

13.1 Complete Code

```
<h1>How to Become a Front End Developer</h1>



<hr />

<h2>Who is a Front End Developer?</h2>
<p>A Front End Developer is responsible for ... </p>
<p>They work with HTML, CSS, and JavaScript ... </p>

<h2>Skills Required</h2>
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
  <li>TypeScript</li>
  <li>Tailwind CSS</li>
  <li>Git & GitHub</li>
</ul>

<h2>Step by Step Guide to Become a Front End Developer</h2>
<ol>
  <li>Master HTML</li>
  <li>Build Static Websites with CSS</li>
  <li>Learn JavaScript Deeply</li>
  <li>Practice with Real Projects</li>
  <li>Learn a Framework (React, Vue, Angular)</li>
</ol>

<a href="https://coderarmy.in" target="_blank">
  Buy the Course to Become a Front End Developer
</a>
```

TAKEAWAY

Is project ne aapko dikhaya ki kaise basic tags ko milakar ek professional-looking page banaya ja sakta hai. **Important:** Aap same content ko alag-alag tarike se structure kar sakte ho — jaise do paragraphs ko ek <p> mein bhi likh sakte ho bas
 se break karke. Choice aapki hai.

14.0 Best Practices

BEST PRACTICE

- 1. Semantic Heading Hierarchy Follow Karo:** H1 → H2 → H3. Jump mat karo. SEO aur screen readers dono ke liye important hai.
- 2. Sirf Ek H1 Per Page:** Multiple H1 se error nahi aayega lekin recommended nahi hai.
- 3. Alt Text Hamesha Do:** Images ke liye meaningful alt attribute likhna mat bhoolo.
- 4. Self-Closing Tags Mein Slash:** `<br />` aur `<hr />` ki tarah slash use karo.
- 5. Comments Use Karo:** Complex sections ko explain karne ke liye comments likho.
- 6. Auto Save On Rakho:** Baar-baar manual save karne se bachne ke liye.

WARNING

HTML ki Khasiyat: Agar aap koi tag galat bhi likh do, toh bhi code chal jaayega. Browser error nahi dega. Lekin ye matlab nahi ki aap careless ho jao. Semantic structure follow karna hamesha best practice hai.

15.0 Quick Revision Checklist

- HTML ka full form — Hyper Text Markup Language
- Hyper Text = Hyperlinks se navigation
- Markup Language = Semantic structure dena
- VS Code extensions — Live Server, Live Preview, Prettier
- index.html = default web page file
- Tags semantic meaning dete hain — structure create karte hain
- 6 Heading levels — H1 (most important) se H6 (least important)
- Ek page mein sirf ek H1 hona chahiye
- Heading hierarchy: H1 → H2 → H3, jump nahi karna
- SEO mein headings critical role play karti hain
- Tag vs Element — Tag sirf symbol, Element poora unit
- HR = horizontal line, BR = line break (don self-closing)
- OL = numbered list, UL = bullet list, LI = list item
- Anchor tag — href (URL), target="_blank" (new tab)
- Image tag — src, height, width, alt (self-closing)
- Alt text accessibility ke liye mandatory hai
- Comments — `<!-- text -->`, output mein nahi dikhte
- Lorem ipsum — dummy text generate karne ke liye